

Package #114

Implement a comprehensive monitoring system that combines acoustic sensing to detect lead dislodgement with thermal monitoring to detect tissue interface changes

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'adverse_event_without_identified_device_' for pacemaker lead: 16 MAUDE reports across 7 distinct devices (20220222 to 20230812); root-cause evidence from 1 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 1 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 4 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, injury, intervention, introducing, lead, maintain, matching, modes, must, problem, producing, recurring, report

Mechanism: A lifecycle-based risk management framework that integrates complication prevention across the entire continuum of pacemaker therapy

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.